

Warm-up

(1) Let $\sum_{n=0}^{\infty} c_n(x-2)^n$ be a power series, and suppose applying the ratio test gives a limit of $L = \frac{1}{4}|x-2|$. What is the radius of convergence?

(2) Let $f(x) = \sum_{n=0}^{\infty} \frac{n+1}{n!} x^n$. Find an expression for $\int f(x) dx$.

(3) Write a power series for $\frac{1}{1-4x}$ centered at 0. What is the radius of convergence?

Medium

(1) Write a power series for $\ln(1+x^2)$ centered at 0. What is the radius of convergence?

For each of the following power series, find the radius and interval of convergence.

$$(2) \sum_{n=0}^{\infty} \frac{n+1}{3^n} (x-1)^n, \quad (3) \sum_{n=0}^{\infty} \frac{n!}{5^n} (x-3)^n, \quad (4) \sum_{n=0}^{\infty} \frac{5^n}{n!} (x-3)^n, \quad (5) \sum_{n=1}^{\infty} \frac{(4x-3)^n}{\sqrt{n}}.$$

Stretch

(1) Find the radius of convergence and the interval of convergence for

$$\sum_{n=1}^{\infty} \frac{n^2 x^n}{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)}.$$

(2) Suppose f is an even function (that is, $f(x)$ and $f(-x)$ are the same) with a power series expression $\sum_{n=0}^{\infty} c_n x^n$ centered at zero. What must the odd coefficients be?